

Yiunfan Hu

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EDUCATION

BSc. Mechanical Engineering at Stony Brook University University Scholar Dean’s list (all semesters) GPA: 3.98 U4 Standing Upper Division MEC Courses: [364] <i>Introduction to Fluid Mechanics</i> , [363] <i>Mechanics of Solids</i> , [301] <i>Thermodynamics</i> , [325] <i>Manufacturing Processes</i> , [316] <i>Instrumentation and Solid Mechanics Lab</i> , [310] <i>Introduction to Machine Design</i> * <i>In progress Spring 2026</i> : [410] <i>Design of Machine Elements</i> , [305] <i>Heat and Mass Transfer</i> , [320] <i>Numerical Methods in Engineering Design and Analysis</i> , [317] <i>Thermal Sciences and Fluid Mechanics Lab</i>	Stony Brook, NY Jun 2024–May 2027 (expected)
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SKILLS

CAD	Autodesk Inventor, Autodesk Fusion, Onshape, SolidWorks
Mechanical	DFM, Rapid Prototyping, 3D Printing, CNC Machining, Manual Machining, Laser Cutting, Plasma Cutting
Electrical	Arduino, Soldering
Software	C, C#, HTML/CSS/JS, MATLAB, Python

EXPERIENCE

Reactive Collision UAV <i>Research Member</i> at Prof. William Stewart’s Soft Flyers Group - Assisted in designing a collision-resistant fixed-wing quadrotor with spring-loaded pivoting wings. - Developed a launcher capable of accelerating the 1 kg UAV to 20 m/s. - Fabricated major components such as the wings using a CNC router.	Stony Brook, NY Jun 2025–present
Stony Brook Robotics Team – MATE ROV <i>Float Team Lead</i> - Design and build a profiling float for the 2026 Competition in Newfoundland, Canada - Developed a hydraulic actuator system and contributed to arm design on the main ROV	Stony Brook, NY Sept 2025–present
Shape Memory Alloy (SMA) Springs <i>Research Member</i> in a joint project with <i>Interacting Robotic Systems Laboratory</i> , <i>Soft Flyers Group</i> , and <i>Yao Research Group</i> - Developed 2-DOF joint with Arduino control system. Poster presentation in Summer Research Symposium at Stony Brook, August 2024. - Designed and fabricated a test platform to control SMA-spring heating and quantitatively measure resulting displacement and temperature change. - Experimentally validated theoretical models from literature and extended simple single-spring models to novel multi-spring networks.	Stony Brook, NY May 2024–present

PROFESSIONAL TRAINING

NASA L/SPACE Proposal Writing and Evaluation Experience Academy <i>Participant – Team Role: Engineer</i> - Workforce preparation academy dedicated in writing and evaluating professional technical proposals taught by NASA scientists, engineers, and managers. - Developed a proposal focused on the laser ignition system for a rotating detonation rocket engine. - Served on a review panel for NASA solicitation responses and gained hands-on experience in evaluating proposals based on solicitation rubric and identifying strengths and weaknesses.	Virtual May–Aug 2025
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EXTRACURRICULAR PROJECTS

Combat Robotics at National Havoc Robot League (NHRL) <i>Team Captain</i> - Leading a team building combat robots in the 12lb-weight-class at NHRL.	Norwalk, CT Jun 2023–present (Jun 2023–Jun 2024)
12lb Vertical Spinner Robot - Designed and built a combat robot from scratch over 4 months with a hard competition deadline. - Obtained rich experience in system integration, control systems, and metal manufacturing. - Competed in September 2023 and June 2024.	
12lb Differential Swerve Drive Robot - Developed a novel compact drive unit enabling omni-directional movement for small combat robots. - Currently designing a robot to integrate the drive units into, expected to compete in 2027.	(Jun 2024–present)
“Javelin” Compact Electric Scooter <i>Solo Maker</i> - Designed and built one of the most compact electric scooters over 6 months under a \$1,000 budget. - Optimized packaging and maximized space utilization in design. - Built a custom 10s2p li-ion battery pack with a 10Ah capacity as an upgrade to the commercial battery pack. - Currently in use for daily commute.	Stony Brook, NY Oct 2024–present